

AI built it. Growth broke it. We rebuilt it.

An AI education platform serving 420,000+ educators

THE STAKES

The platform began as an AI-generated, no-code-style implementation. It worked — too well: growth outran what that architecture could ever carry, and neither the founders nor the app-builder tooling could scale it. Every week at that wall is churn and lost subscriptions. This is the story most companies shipping AI-built software are about to live: it works until it matters.

WHAT WE DID

The company brought in a team to rebuild it properly — the backend was **two engineers**. The rebuilt platform shipped as the public 2.0 launch with 150+ AI tools:

- **Production LLM features that survive real traffic:** RAG with LangChain and vector embeddings, multi-provider routing (Azure OpenAI, AWS Bedrock), streaming endpoints
- **The revenue plumbing:** subscriptions, school management, Stripe billing, Cognito authentication
- **The scaling work:** sync-to-async migration (ASGI/Uvicorn), AWS ECS deployments, CI/CD
- **An email system admins run without engineers:** templates with live HTML preview and automated workflows, designed and built end to end

OUTCOMES

- The rebuilt platform launched publicly and serves **420,000+ educators**.
- An AI-generated prototype became a scalable production system — without the business stopping.
- We know exactly where AI-generated codebases break at scale, because we have rebuilt one that 420,000 people depend on.

WHY IT MATTERS

If your product was built fast with AI tooling and is now meeting real traffic, this is the playbook: keep the product live, rebuild the foundations underneath it, ship the launch on time.

STACK

Python · Django · LangChain · Azure OpenAI · AWS Bedrock · AWS ECS · Cognito · Stripe

This is the engineering standard behind Alpine's fixed-price delivery audit: findings with file-and-line evidence, an implementation playbook, and an execution-ready backlog your team can run with. References on request.

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